Week 2 Progress Report

Road Pothole Detection System Using YOLO-Based Deep Learning

Programme: B.Tech CSE (Artificial Intelligence & Machine Learning)
UPES, Dehradun
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Team Members & Responsibilities

Roll No.	Name	Responsibility
500119453	**Aditya Khatri**	Model training, running inference pipeline, performance analysis
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1. Week 2 Overview

The project made great progress in week two. This week was devoted to finishing the model training and doing a thorough inference evaluation, building on the design and pipeline setup from Week 1. By the conclusion of the week, the inference pipeline had been tested for all three input modes (image, video, and webcam), the YOLOv12n model had been trained on the Kaggle Pothole Dataset, and a performance analysis using specific metrics had been finished. The outcomes are positive and provide a solid foundation for improvement in the upcoming weeks.

2. Model and Dataset Details

The YOLOv12n model, which is the nano version of the YOLOv12 architecture, was selected for this training run. It was chosen because of its attention-centric architecture, particularly the A2 and R-ELAN modules, which effectively balance computational efficiency and detection accuracy. This makes it ideal for real-world deployment scenarios where inference must operate on edge devices or consumer-grade hardware without significantly compromising accuracy.

The training dataset, which includes 665 photos of a single class—potholes—was obtained from Kaggle. Despite its small size, the dataset is well-labeled and covers a respectable range of road conditions. The trained best weights are stored under runs/detect/yolov12n_pothole_detector5/weights/best.pt. The model was trained using transfer learning from the pre-trained yolov12n.pt weights.

Component	Detail
Model Architecture	YOLOv12n (Nano Variant)
Trained Weights	runs/detect/yolov12n_pothole_detector5/weights/best.pt
Dataset Source	Kaggle Pothole Dataset
Dataset Size	665 images
Classes Detected	1 — Pothole
Key Architecture Feature	Attention-Centric Design (A2 and R-ELAN) for efficiency

3. Inference Pipeline

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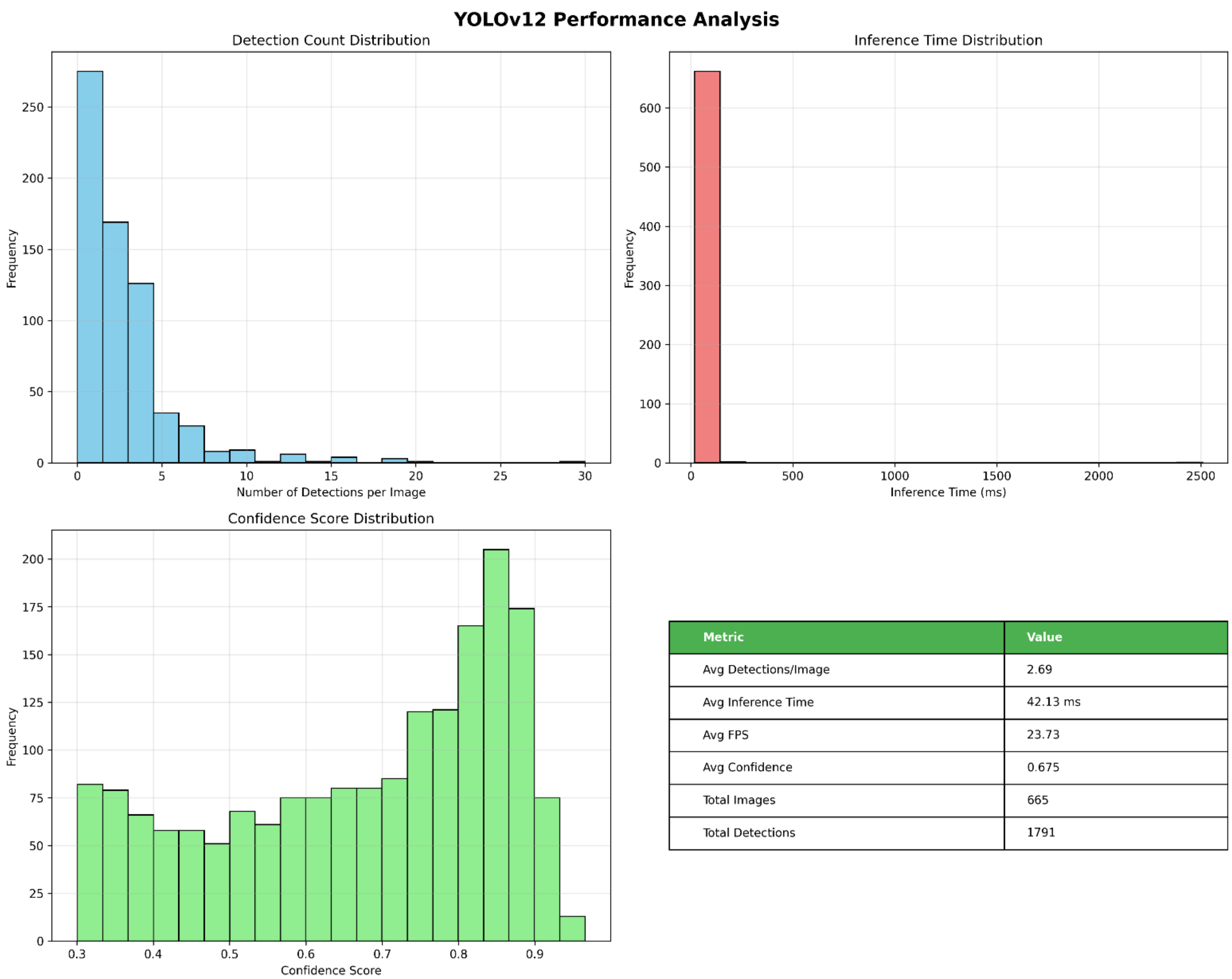
Input Mode	Setting	Output
Video Processing	"input_video.mp4"	Annotated video saved as pothole_detection_result.mp4
Live Webcam	"0"	Real-time detection display (no file saved)
Image Processing	"input_image.png"	Annotated image saved to runs/ directory

The proper operation of each of the three modes was confirmed. The output video with annotations (pothole_detection_result).mp4) was produced from the input test video and verified to display bounding boxes with confidence scores superimposed around identified potholes. A pothole was accurately identified with a confidence score of 0.88 in an example detection via visual inference, which is a good outcome for a first training run.

4. Performance Analysis

A comprehensive performance analysis was run on the trained YOLOv12n model across the full dataset of 665 images. The results provide a clear picture of where the model currently stands and what areas might need attention in the next training iteration.

Metric	Value
Total Images Processed	665
Total Detections	1,791
Average Detections per Image	2.69
Average Inference Time	42.13 ms
Average FPS	23.73
Average Confidence Score	0.675



4.1 Inference Speed

An average inference time of 42.13 ms per image, or around 23.73 frames per second, was attained using the model. This is encouraging for a first training run and very near to the 25 FPS target specified in the project specifications. It should be possible to regularly reach the 25 FPS target on mid-range hardware with additional optimization, such as applying model quantization or significantly lowering input resolution.

4.2 Confidence Score Distribution

A significant percentage of detections lie between 0.75 and 0.95, according to the confidence score distribution, which peaks between 0.85 and 0.90. Some lower-confidence detections in the 0.30 to 0.50 range drag down the average confidence across all detections, which is 0.675. These lower-confidence detections are worth looking into because they could be false positives on road surface textures that resemble potholes, heavily occluded instances, or partial potholes.

4.3 Detection Count Distribution

The majority of the dataset's photos have between 0 and 5 detections, which is in line with what would be expected from road imagery; most frames have a few potholes, while a tiny number of frames have damage clusters. A few photos have abnormally high detection counts (15 or higher), which could mean the model is over-detecting under specific circumstances like dim lighting or damaged road surfaces.

5. Project Structure

The project directory is now fully organised with clearly separated folders for each stage of the pipeline. The structure at the end of Week 2 is as follows:

Folder / File	Contents
data/	Kaggle dataset images and YOLO format labels
outputs/	Detection results — image.jpg, input_image.jpg, pothole_detection_result.mp4
runs/	Trained model weights and inference outputs from each run
yolo_comprehensive_results/	Performance analysis charts and metrics
train_yolov12.py	Script used for model training
run_detector.py	Inference script — handles image, video and webcam input
convert_pothole_to_yolo.py	Annotation format conversion utility
pothole_dataset.yaml	Dataset paths and class configuration

REPORT.txt	Model summary and performance report
yolo11n.pt / yolov12n.pt	Pre-trained base weights for both model variants

6. Challenges Encountered

The aim of 25 frames per second is slightly exceeded by the average FPS of 23.73. This suggests that considerable optimization will be required before the system can be accurately classified as real-time, even though it is not a significant worry at this time. This will be examined in Week 3, perhaps via inference batch tuning or input resolution correction.
It is worthwhile to look at the higher detection counts on a small group of photos. While some of them might be true instances of several potholes appearing in a single frame, others might be false positives brought on by the model reacting to puddles, shadows, or road surface textures that resemble potholes visually. These high-count frames will be manually reviewed.

Data collecting on local roads has not yet begun. This was supposed to happen in Week 2, but finishing the training run and evaluation took precedence. It is still scheduled for Week 3 and will be crucial in determining how well the model applies to Indian road conditions in particular.

7. Plan for Week 3

Improving the model and expanding the dataset will be the primary goals for Week 3. The next step is to add more data, especially from the RDD2022 dataset, to expand the training set size and enhance generalization now that a working baseline with YOLOv11n and initial YOLOv12n results has been established. This week, there will also be an attempt to gather local road footage.

After both models have been trained under the same conditions, a formal comparison between YOLOv11n and YOLOv12n will be conducted. The mAP, precision, recall, inference speed, and confidence distributions will all be compared side by side. The outcomes will be thoroughly recorded.

Depending on the outcome, some hyperparameter tuning may also be carried out, such as changing the augmentation settings to solve the over-detection problem seen in some frames, testing with different input resolutions, or adjusting the confidence threshold.

8. References

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